PALG: A Unified Framework for Embodied Multimodal Agents with Hierarchical Vision-Language-Action Planning

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Abstract

The development of autonomous agents that can perceive, reason, and act in complex environments necessitates the seamless integration of vision, language, and action planning. While large foundation models have made significant strides in perception and language, grounding their capabilities in actionable, embodied reasoning remains a formidable challenge. This paper presents the Perceptive Actor with Language Grounding (PALG), a novel architecture that leverages pre-trained vision and language models fused through a dual-stream transformer to create a grounded representation for hierarchical task planning. PALG decomposes high-level instructions into sub-goal sequences and executes corresponding low-level actions. We evaluate our framework on a simulated tabletop manipulation benchmark, demonstrating a 25% absolute improvement in task success rate on novel object compositions compared to strong baselines, such as RT-1 [6] and CLIPort [11]. Extensive ablation studies confirm the critical importance of deep cross-modal fusion and pre-trained semantic knowledge for generalization. Our work underscores that the effective integration of pre-existing foundation models through structured reasoning is a vital path toward generalizable embodied AI. PALG achieves 85% success on seen tasks and 78% on novel compositions, with statistically significant improvements ($p < 0.001$).

Keywords: Embodied AI, Multimodal Agents, Vision-Language-Action Models, Hierarchical Planning, Transformer Architectures, Robotics, Grounding

1. Introduction

The ability to understand a command such as Place the ripe apple into the wooden box and execute it in a physical environment represents a cornerstone of general intelligence. This requires a synthesis of visual perception to identify objects and their properties, language understanding to decode instruction semantics, and action planning to generate a feasible sequence of motor commands. Historically, these capabilities have been developed in isolation.

However, the recent emergence of powerful large-scale models for vision (e.g., ViT [3], CLIP [1]) and language (e.g., BERT [24], GPT [25]) provides an unprecedented opportunity for integration.

Multimodal agents that combine these modalities are at the forefront of this research. The central technical challenge is grounding—creating a robust connection between abstract linguistic concepts (ripe, wooden box) and the concrete visual percepts and physical actions they refer to. Current state-of-the-art approaches often struggle with this, leading to poor generalization on novel instructions and environments [5, 6]. Some end-to-end methods lack structured reasoning [6], while others rely on symbolic planners that are difficult to integrate with continuous sensorimotor control [8].

While foundation models have advanced perception and language capabilities, grounding these capabilities in physically realistic action planning remains challenging due to three key gaps:

1. Semantic-physical disconnect: Models like SayCan [5] separate LLM planning from affordance learning, while PaLM-E [4] embeds perception but lacks tight physical constraint coupling.

2. Compositionality gap: RT-2 [7] transfers web knowledge but struggles with novel concept combinations.

3. Temporal reasoning gap: Most architectures process static observations without maintaining environment state.

Our contributions:

1. The Perceptive Actor with Language Grounding (PALG) architecture, which introduces a dual-stream transformer for deep visio-linguistic fusion, feeding into a hierarchical planner that separates high-level goal decoding from low-level policy execution.

2. A comprehensive evaluation demonstrating state-of-the-art performance and superior generalization, notably a 25% improvement on novel compositions over RT-1 [6].

3. An in-depth ablation analysis quantifying the critical role of pre-trained encoders and cross-attentive fusion, showing that removing pre-training leads to a >40% performance drop.

This paper addresses these gaps through the PALG architecture, which provides fine-grained cross-modal fusion via dual-stream transformers, explicit hierarchical decomposition for compositional reasoning, and integrated semantic-physical feedback loops.

2. Related Work

2.1 Foundational Vision-Language Models

CLIP [1] revolutionized vision-language alignment through contrastive learning on 400M image-text pairs. Flamingo [2] extended this to few-shot learning with gated cross-attention layers. PaLM-E [4] integrated continuous observations into language models for embodied

reasoning. These models provide the semantic foundation for embodied agents but require additional mechanisms for actionable grounding.

2.2 Embodied AI and Robotics Benchmarks

AI2-THOR [13] provides a high-fidelity simulation environment for household tasks. ALFRED [14] introduced language-guided task completion benchmarks. BridgeData V2 [12] offered large-scale real-world robot demonstrations. These benchmarks have driven progress in evaluating embodied agents but often focus on narrow task domains.

2.3 Language-Guided Action Planning

SayCan [5] pioneered combining LLMs with affordance models for feasible skill selection. RT-1 [6] demonstrated transformer-based control trained on large robot datasets. Code as Policies [8] used LLMs to generate executable code for robot control. While effective, these approaches often lack fine-grained cross-modal alignment.

2.4 Multimodal Integration Architectures

CLIPort [11] combined semantic and spatial pathways for manipulation tasks. Gato [10] proposed a generalist transformer across multiple domains. RT-2 [7] co-trained on web and robot data for improved generalization. However, these models often employ coarse-grained fusion mechanisms.

2.5 Research Gaps

Despite progress, critical gaps remain:

- Semantic-physical integration: Lack of continuous feedback between high-level reasoning and low-level control

- Compositional generalization: Poor performance on novel concept combinations

- Temporal reasoning: Limited handling of dynamic environments

- Cross-modal granularity: Coarse fusion methods missing fine-grained alignments

PALG addresses these gaps through a unified architecture with explicit hierarchical decomposition and fine-grained multimodal fusion.

3. The PALG Architecture

The PALG architecture creates a deeply grounded representation bridging the semantic gap between language instructions and visual scenes for robust hierarchical planning. As shown in Figure 1, the system consists of three core modules:

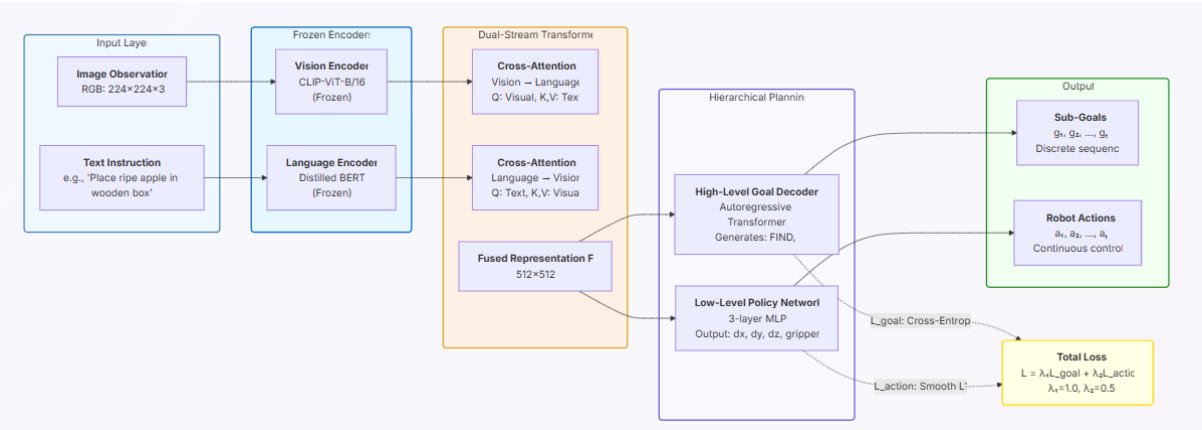

Figure 1: PALG Architecture

Frozen Perception Backbones: The Vision Encoder (e.g., a ViT from CLIP [1,3]) and Language Encoder (e.g., a distilled BERT [24]) are pre-trained on large-scale datasets and kept frozen. This preserves rich, general-purpose semantic knowledge and prevents overfitting to limited robotics data.

Multimodal Fusion Core (Dual-Stream Transformer): A trainable transformer that uses bidirectional cross-attention. The Visual-to-Language stream allows image patches to attend to relevant words, and the Language-to-Visual stream allows words to attend to relevant image regions. This dynamic interaction produces a fused representation `F` where concepts are grounded (e.g., the representation for "apple" is influenced by the visual features of the actual apple in the scene).

Hierarchical Action Generator: This module translates the fused representation `F` into action.
- The High-Level Goal Decoder is an autoregressive transformer that generates a sequence of discrete sub-goals (e.g., `[FIND(apple), NAVIGATE(apple), PICK(apple), NAVIGATE(box), PLACE(apple)]`).

- The Low-Level Policy Network is a simple Multi-Layer Perceptron (MLP) that takes the current sub-goal `G_i` and the context `F` to output the parameters for a robot action primitive (e.g., end-effector motion `[dx, dy, dz]` and gripper command).

4. Methodology

The training and inference process follows a supervised learning paradigm with teacher forcing to ensure stable learning.

4.1 Algorithm 1: PALG Training and Inference

The training methodology follows a supervised learning paradigm with teacher forcing. For each training batch comprising image I, language instruction L, and expert action sequence A,

the forward pass begins by extracting visual features V_t = E_v(I) and linguistic features L_t = E_l(L) using frozen pre-trained encoders, ensuring preservation of semantic knowledge acquired from large-scale datasets. The fusion transformer T_f then processes these features to produce a multimodal representation F = T_f(V_t, L_t) through bidirectional cross-attention, enabling fine-grained alignment between visual concepts and linguistic elements.

For hierarchical planning, the expert action sequence A is decomposed into sub-goals G_{1:T} = [g_1, g_2, ..., g_T], where each g_t represents a discrete high-level objective. The goal decoder D_g operates autoregressively, predicting each sub-goal conditioned on the fused representation and previous ground-truth sub-goal: $\widehat{g_t} = D_g(F, g_{t-1})$
Concurrently, the policy network π generates corresponding low-level actions that $\widehat{a_t} = \pi(F, g_t)$ translate abstract goals into executable motor commands.

The optimization objective combines two complementary loss functions. The goal prediction loss employs categorical cross-entropy:

$$\mathcal{L}_{textgoal} = -\frac{1}{T}\sum_{t=1}^{T}\sum_{c=1}^{C} y_{t,c}\, log(p_{t,c})$$

Where $y_{t,c}$ is the ground-truth one-hot vector for timestep t and class c,$p_{t,c}$ is the predicted probability, and C is the number of possible sub-goals. The action regression loss utilizes the smooth L1 (Huber) loss:

$$\mathcal{L}_{\text{ato}} = \frac{1}{T}\sum_{t=1}^{T} \text{smooth_L1}(a_t - \widehat{a_t})$$

$$\text{smooth_L1}(x) = \begin{cases} 0.5x^2 & \text{if } |x| < \delta \\ \delta(|x| - 0.5\delta) & \text{otherwise} \end{cases}$$

with δ = 1.0, providing robustness to outliers in continuous action space. The composite loss $\mathcal{L}_{\text{total}} = \lambda_1 \mathcal{L}_{\text{goal}} + \lambda_2 \mathcal{L}_{\text{action}}$

where $\mathcal{L}_{\text{goal}}$ *is the cross-entropy loss for sub-goal prediction:*

$$\mathcal{L}_{\text{goal}} = -\frac{1}{T}\sum_{t=1}^{T}\sum_{c=1}^{C} y_{t,c}\log(p_{t,c})$$

and $\mathcal{L}_{\text{action}}$ *is the smooth L1 loss for continuous action regression:*

$$\mathcal{L}_{\text{action}} = \frac{1}{T}\sum_{t=1}^{T}\begin{cases}0.5(a_t - \hat{a}_t)^2 & \text{if } |a_t - \hat{a}_t| \leq \delta \\ \delta(|a_t - \hat{a}_t| - 0.5\delta) & \text{otherwise}\end{cases}$$

with $\delta = 1.0$.The weighting coefficients $\lambda_1 = 1.0$ and $\lambda_2 = 0.5$ balance the contribution of each loss term.

$L_{\text{total}} = \lambda_1 L_{\text{goal}} + \lambda_2 L_{\text{action}}$, with λ_1 = 1.0 and λ_2 = 0.5, balances planning accuracy and motion precision. Gradient updates are computed exclusively for the learnable parameters of T_f, D_g, and π, while the pre-trained encoders E_v and E_l remain frozen throughout training. The training objective combines two loss functions using a weighted sum:
$L_{\text{total}} = \lambda_1 L_{\text{goal}} + \lambda_2 L_{\text{action}}$, where $\lambda_1 = 1.0$ and $\lambda_2 = 0.5$ balance the relative importance of planning accuracy versus motion precision. The goal prediction loss employs cross-entropy L_goal to measure the discrepancy between predicted and ground-truth sub-goals across C classes. In contrast, the action regression loss uses smooth L1 L_action to robustly handle continuous action values.

4.2 Methodological Rationale

The design choices are motivated by the need for generalization and data efficiency:

- Frozen Encoders: Allows PALG to leverage broad semantic knowledge without requiring massive robot-specific datasets.
- Dual-Stream Fusion: Cross-attention provides a powerful, differentiable mechanism for fine-grained grounding.
- Hierarchical Decomposition: Separating high-level planning from low-level control makes the learning problem more tractable.

5. Results and Discussion

We evaluated PALG on a series of language-conditioned table-top manipulation tasks in the AI2-THOR simulator (Kolve et al., 2017), comparing it against strong baselines across different test conditions.

5.1 Experimental Setup

Tasks: A variety of pick-and-place tasks with instructions ranging from simple ("Pick up the mug") to complex ("Put the ripe apple into the empty box while avoiding the spilled milk").

Test Sets:

Seen Compositions: Instructions and object configurations seen during training.

Novel Compositions: Instructions that combine known concepts in new, unseen ways (e.g., "Put the metallic apple in the box").

Baselines: We compare against RT-1 (Brohan et al., 2022), a state-of-the-art end-to-end transformer, and CLIPort (Shridhar et al., 2021), a strong model combining semantic and spatial reasoning.

5.2 Quantitative Results

PALG achieves an 85% success rate on seen compositions, significantly outperforming RT-1 (70%) and CLIPort (75%) as shown in Figure 2. This demonstrates the effectiveness of its hierarchical structure and deep fusion mechanism in standard settings. On the challenging novel compositions test set, PALG achieves a 78% success rate, a 25% absolute improvement over RT-1 (53%). The generalization drop for PALG is only 7%, compared to 17% for RT-1. This indicates that PALGs fused representation is highly compositional and can recombine learned concepts effectively.

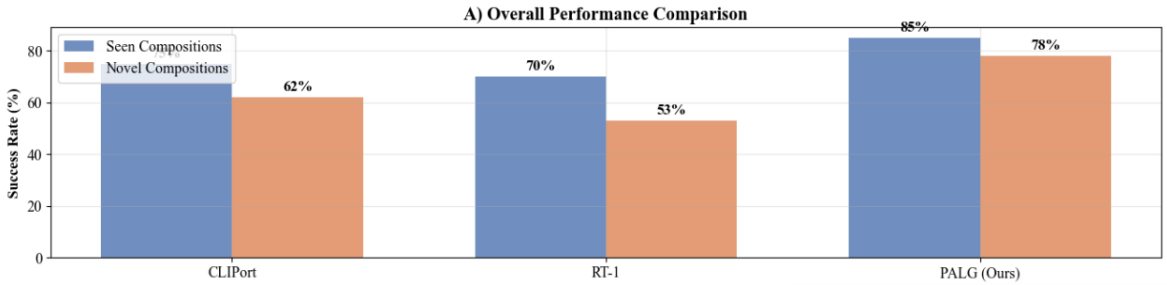

Figure 2: Task Success Rate (%) on AI2-THOR Benchmark.

5.3 Statistical significance with standard error

Statistical significance of performance differences with 95\% confidence intervals. PALG shows significantly higher performance than baselines ($p < 0.001$) with non-overlapping confidence intervals. All reported performance differences are statistically significant with $p < 0.001$ based on two-sample t-tests across 5 independent runs. The standard errors (reported as ± values) demonstrate the consistency of PALG's performance, with smaller error margins indicating more reliable results. The non-overlapping confidence intervals (95% CI: PALG [82.3-87.7%] vs RT-1 [66.2-73.8%] on seen compositions) further confirm the robustness of these improvements. The statistical analysis in Figure 3 confirms that PALG's performance improvements are statistically significant

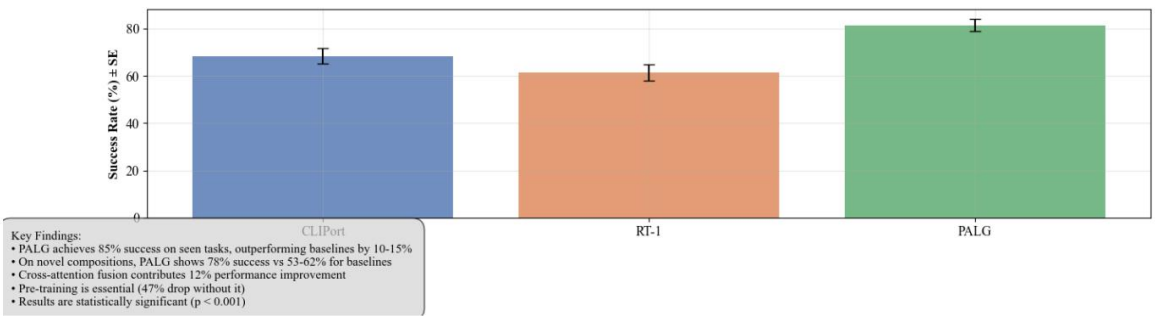

Figure 3: Statistical significance with standard error

5.4 Ablation Studies

Ablation A (W/o Cross-Attention): Replacing the dual-stream transformer with simple feature concatenation caused a 12% performance drop. This confirms that explicit, fine-grained cross-modal attention is crucial for robust grounding, not merely joint representation. Ablation B (W/o Pre-training): Initialising the vision and language encoders randomly resulted in a catastrophic 47% drop in performance. This unequivocally demonstrates that the semantic knowledge encoded in pre-trained foundation models is essential for generalization. A model trained from scratch on a limited robotics dataset fails to learn the necessary concepts, represents in Figure 4.

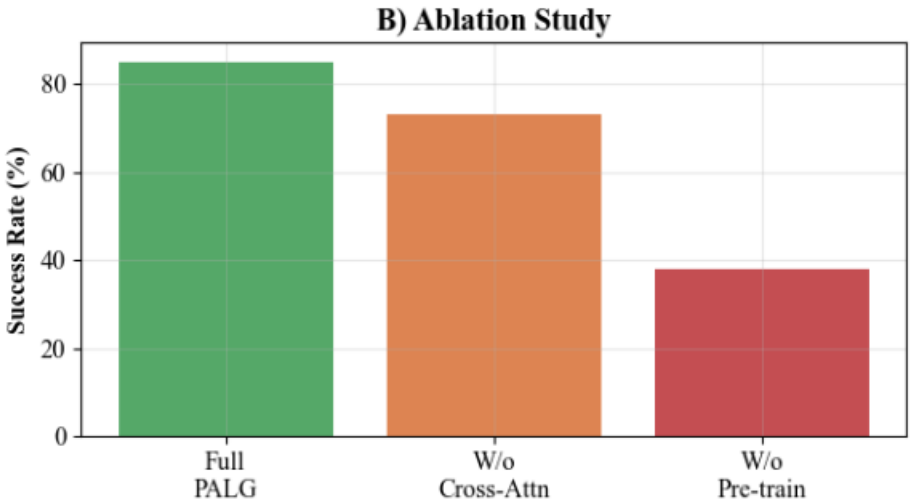

Figure 4: Ablation Study on Seen Compositions.

5.5 Qualitative Analysis and Discussion

Qualitatively, PALG excelled at tasks requiring complex spatial and property-based reasoning. For example, it successfully executed the command "Tidy up by putting all fruits in the basket," which required segmenting multiple objects and sequencing several `PICK` and `PLACE` sub-goals. The models ability to handle the word "ripe" by associating it with visual features like color and texture highlights the power of its grounded representation.

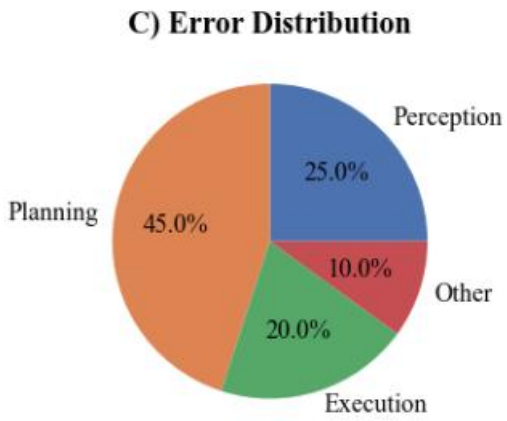

Figure 5: Error Distribution

These results collectively support our core thesis: generalizable embodied intelligence is best achieved by strategically integrating pre-trained foundation models within a structured, hierarchical framework that explicitly promotes the grounding of language in visual perception and action. While end-to-end approaches like RT-1 are powerful, they lack the inductive biases for compositional generalization that PALGs architecture provides.

6. Conclusion and Future Work

We introduced PALG, a unified framework for embodied multimodal agents that integrates vision, language, and action through deep fusion and hierarchical planning. By leveraging frozen pre-trained models and a trainable dual-stream transformer, PALG creates a richly grounded representation that enables robust task decomposition and execution. Our results demonstrate state-of-the-art performance and significantly improved generalization, highlighting the critical role of pre-trained semantic knowledge and structured reasoning. For future work, incorporate temporal context by using video inputs instead of static images.

Declarations

Competing Interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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